Juan Antonio Monleón de la Lluvia

Nuclear Engineer · PhD in Nuclear Physics

CONTACT

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EDUCATION

PhD in Nuclear Physics
Université Paris-Saclay · ED PHENIICS — 2023-2026

Double Master in Industrial Engineering and Nuclear Science and Technology
Polytechnic University of Madrid — 2021-2023

Master in Management and Direction of SMEs
EFEM Business School — 2024-2025

Bachelor in Industrial Technology Engineering
Polytechnic University of Madrid — 2017-2021

TECHNICAL SKILLS

Nuclear Codes: MCNP, Serpent, OpenMC, NJOY, ADVANTG

Programming: Python, Git, AI-assisted development

Documentation: LaTeX, Microsoft 365

ABOUT ME

Nuclear engineer with a PhD in Nuclear Physics from Université Paris-Saclay, working on nuclear data and on the sensitivity and uncertainty analysis of neutronic simulations, from the evaluation of covariances to their propagation in shielding and reactor-dosimetry calculations. Proactive and quick to adapt to new codes, tools and problem domains.

RESEARCH EXPERIENCE

Doctoral Research in Nuclear Data & Neutronics

ASNR, France · Université Paris-Saclay — Nov 2023 - Oct 2026

Thesis: **"Nuclear-Data Uncertainty Quantification for PWR Vessel Aging with a ^{56}Fe Elastic-Scattering Evaluation"** *(defended 1 October 2026)*

- Sensitivity analysis and Monte Carlo sampling to propagate nuclear-data uncertainties; re-evaluated ^{56}Fe elastic angular distributions and covariances from EXFOR data
- Fast-neutron fluence calculations for the PWR reactor vessel with MCNP, using variance reduction (ADVANTG weight windows) for deep-penetration transport
- Supervised a Master's student (Mar–Sep 2025) on nuclear-data uncertainty quantification in the PWR source term, the input to the vessel fluence calculations

Master's Thesis Research (MII)

UPM, Spain — Jun 2023 - Oct 2023

- Developed **"Enhancing the EXFOR nuclear data library using Machine Learning Techniques"**
- Implemented outlier detection methods for proton-induced reactions

Research Internship — Master's Thesis (MUCTN)

SCK CEN, Belgium — Sep 2022 - Jun 2023

- Completed Master's thesis titled **"Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques"**
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

Master Internship

ENUSA Advanced Industries, Spain — Oct 2021 - Apr 2022

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

LANGUAGES

Spanish Native language

English C2 Level

French C1 Level

AWARDS

Best Academic Record, Nuclear Master
Polytechnic University of Madrid (2022-2023)

Best Master Thesis Award
SCK CEN, Belgium (2023)

Best Poster Award
Journées des Thèses of IRSN (2024)

COMMUNITY

JEFF Community
Active member · Joint Evaluated Fission and Fusion File (OECD/NEA)

INDEN Community
Member · International Nuclear Data Evaluation Network (IAEA)

SINBAD Community
Member · Shielding Integral Benchmark Archive & Database (OECD/NEA)

REFERENCES

Dr. Mariya Brovchenko
ASNR, France
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Dr. Dimitri Rochman
Nagra, Switzerland
dimitri.rochman@nagra.ch

Prof. Óscar Cabellos
Universidad Politécnica de Madrid, Spain
oscar.cabellos@upm.es

PUBLICATIONS

- Y. Celik, **J. A. Monleon de la Lluvia**, G. Van den Eynde, "Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6", *Nuclear Science and Engineering (under review)* — 2026
- **J. A. Monleon de la Lluvia**, M. Brovchenko, D. Rochman, E. Dumonteil, "Toward Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations", *Nuclear Science and Engineering* 200(2), 257–279. DOI: 10.1080/00295639.2025.2510048 · HAL — 2025

CONFERENCES & PRESENTATIONS

- **J. A. Monleon de la Lluvia**, M. Brovchenko, D. Rochman, "Evaluation of elastic angular distribution covariances for ^{56}Fe from EXFOR experimental data" — WONDER 2026, Aix-en-Provence (oral) — Jun 2026
- **J. A. Monleon de la Lluvia**, M. Brovchenko, D. Rochman, "Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to ^{56}Fe" — PHYSOR 2026, Turin (oral) — Apr 2026
- Y. Celik, **J. A. Monleon de la Lluvia**, G. Van den Eynde, "Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6" — PHYSOR 2026, Turin (poster) — Apr 2026
- D. Timpano, D. Rochman, **J. A. Monleon de la Lluvia**, M. Brovchenko, M. Hursin, "Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations" — PHYSOR 2026, Turin — Apr 2026
- **J. A. Monleon de la Lluvia**, M. Brovchenko, D. Rochman, "Nuclear Data Uncertainty Quantification Using the VENUS-3 Benchmark Experiment" — ND 2025, Madrid (oral) — Jun 2025
- **J. A. Monleon de la Lluvia**, M. Brovchenko, D. Rochman, E. Dumonteil, "Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing-like problems" — ANS Winter Conference RPSD, Orlando (oral) — Nov 2024

SOFTWARE & TOOLS

KIKA — Nuclear-Data Processing and Analysis Software

Author & lead developer — 2025 - Present

- Desktop application for nuclear-data processing and analysis: ENDF, ACE, EXFOR and SDF data, NJOY processing, MCNP and Serpent outputs, and nuclear-data sampling, sensitivity and uncertainty analysis
- Computational core in **kika-nd**, an open-source Python library (PyPI, BSD-3-Clause)

TRAINING

- FUDGE/(MC)GIDI/GNDS Course — NEA (online) — Oct 2026
- OpenMC Course — NEA, Paris — Dec 2025
- NJOY Course — NEA, Paris — Dec 2024
- Sensitivity Analysis Summer School (SAMO) — UNIPR, Parma — Jun 2024
- Serpent Bootcamp — NEA, Paris — Nov 2023
- Course in "Nuclear Data for Energy and Non-Energy Applications" — GREAT PIONEER — Dec 2022